

INTRODUCTION TO OPERATING SYSTEM

An operating system(os) is the software component of a computer system that is responsible for the management and coordination of activities and sharing of the resources of the computer . The OS acts as a host for application programs that are run on the machine. It acts as an intermediary between the user of a computer and computer hardware .

OS offer a number of services to application programs and users. Application access these services through application programming interfaces (APIs) or system calls. By using these interfaces ,the application can request a service from the os ,pass parameters ,and receive the results of the operations.

SCHEDULING ALGORITHMS

A] PROCESS SCHEDULING:

A process scheduler schedules different processes to be assigned to the CPU based on particular scheduling algorithms . These are six popular process scheduling algorithms:

- * First-Come , First-Served (FCFS) scheduling
- * Shortest-Job-Next (SJN) Scheduling
- * Priority Scheduling
- * Shortest Remaining Time
- * Round Robin(RR) Scheduling
- * Multiple-Level Queues Scheduling

B] DISC SCHEDULING:

Disk scheduling is done by operating systems to schedule I/O requests arriving for the disk. Disk scheduling is also known as I/O scheduling

Disk scheduling is important because :

- * Multiple I/O requests may arrive by different processes and only one I/O request can be served at a time by the disk controller .
- * Two or more requests may far from each other so can result in greater disk arm movement
- * Harddrives are one of the slowest part of the computer system and thus need to be accessed in an efficient manner

MEMORY MANAGEMENT

Memory management is the functionality of an operating system which handles or manages primary memory and moves processes back and forth between main memory and disk during execution .

Memory management is a form of resource management applied to computer memory. The essential requirement of memory management is to provide ways to dynamically allocate portions of memory programs at their request , and free it for reuse when no longer needed.

SEMAPHORE:

A semaphore is a variable or abstract datatype used to control access to a common resource by multiple processes in a current system such as multitasking operating system. A semaphore is simply variable . This variable is used to solve critical section problems and to achieve process synchronization in multi processing environment.

A useful way to think of a semaphore as used in the real-world system is as a record of how many units of a particular resource are available , coupled with operations to adjust that record safely (i.e., to avoid race conditions) as units are acquired or become free , and , if necessary, wait until a unit of the resource become available .

Semaphore is simply a variable that is non-negative and shared between threads . A semaphore is a signal mechanism , and a thread that is waiting on a semaphore can be signaled by another thread . It uses two atomic operations ,1)wait , and 2)signal for the process synchronization.

DEADLOCK:

In curent computing, a deadlock is a state in which each member of a group is waiting for another member , including itself, to take action ,such as sending a message or more commonly releasing a lock. Deadlock is a common problem in multiprocess systems ,parallel computing ,resources and implementing process synchronization.

In an operating system a deadlock occurs when a process or thread enters a waiting state because a requested system resource is held by another waiting process, which in turn is waiting for another resource held by another waiting proces. Deadlock is a situatin where a set of processes are blocked because each process is holding a resource acquired by some other processes.

Deadlock can arise if following four conditions hold simultaneously:

1.Mutual Exclusion:

One or more than one resources are non_sharable (Only one process can use at a time)

2.Hold and Wait:

A process is holding atleast one resources and waiting for resources .

3.No Preemption:

A resource cannot be taken from a process uless the process release the resource.

4.Circular Wait:

A set of processes are waiting for each other in circular form.

Methods of handling dead lock:

There are three ways to handle deadlock:

1) Deadlock prevention or avoidance: The idea is to not let the system into deadlock state. One can zoom into each category individually , Prevention is done by negating one of above mentioned necessary conditions for deadlock.

2) Deadlock detection and recovery : Let deadlock occur , then do preemption to handle it once occurred .

3) Ignore problems all together : If deadlock is very rare ,then let it happen and reboot the system .This is the approach that bothe Windows and Unix take
